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Effective hyperspectral image classification based on segmented PCA and 3D-2D CNN leveraging multibranch feature fusion

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ABSTRACT

We present an innovative hyperspectral image (HSI) classification method addressing challenges posed by closely spaced wavelength bands. Our approach combines 3D-2D convolutional neural networks (CNNs) with multi-branch feature fusion for improved spectral-spatial feature extraction. Using segmented principal component analysis (Seg-PCA), we reduce HSIs' spectral dimensions into global and local intrinsic characteristics. The integration of 3D and 2D CNNs captures joint spectral-spatial features, while a multi-branch network extracts and merges diverse local features along the spectral dimension. Our method outperforms existing approaches, achieving remarkable accuracy of 99.27%, 100%, and 99.99% on Indian Pines, Salinas Scene, and University of Pavia datasets, respectively.

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Convolutional neural networks; remote sensing; hyperspectral image classification; deep learning; segmented PCA; multi-branch 3D-2D CNN; feature fusion

1. Introduction

Remote sensing (RS) hyperspectral images (HSIs) combine imaging and spectroscopy to detect 2D spatial characteristics and 1D spectral information of targets, making them useful in several disciplines, including vegetation analysis, medicine, atmospheric monitoring, precision agriculture, hydrological detection, wind forecasting, and others (Paul *et al*. 2021, Hell *et al*. 2022, Nega *et al*. 2022, Hajaj *et al*. 2023). HSIs have a high spectral resolution and a broad spectral coverage range, comprising detailed information that can be difficult to analyse due to the volume of data. More effectively utilising spatial and spectral information can substantially enhance material detection and object recognition (Hassanzadeh *et al*. 2022, Islam *et al*. 2022, Yang *et al*. 2023).

HSI classification aims to designate each pixel vector to a particular land cover class. Effectively utilising the plethora of spatial and spectral characteristics presents a formidable challenge in HSIs. Despite the prospective information provided by immense spectral bands, the classification accuracy initially increases and then decreases due to the high-dimensional nature of hyperspectral data, which results in the Hughes phenomenon (Hughes 1968). To tackle these difficulties, a common practice in HSI classification involves

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employing feature reduction or dimensionality reduction as a preliminary step. Feature reduction can be achieved by extracting meaningful features and subsequently selecting a top-ranked subset from these features. The extraction of features and the classification of hyperspectral images can be accomplished through both conventional machine learning (ML)-based approaches and deep learning (DL)-based approaches.

1.1. ML-based approach

Traditional ML classification methods, like support vector machines (SVM) (Waske *et al*. 2010, Chang and Lin 2011, Uddin *et al*. 2021a), random forest (RF) (Joelsson *et al*. 2005), K-nearest neighbour (KNN) (Song *et al*. 2016), and artificial neural networks (ANN) (Liang *et al*. 2018), are foundational in machine learning. In general, these techniques depend on handcrafted features and predominantly employ 1-D spectral features. Consequently, dimensionality reduction techniques for the spectral dimension, such as principal component analysis (PCA) (Jia and Richards 1999, Uddin *et al*. 2021b), independent component analysis (ICA) (Wang and Chang 2006), minimum noise fraction (MNF) (Islam *et al*. 2022, Siddiqa *et al*. 2022), and linear discriminant analysis (LDA) (Fabiyi *et al*. 2021, Afjal *et al*. 2023), are receiving increased attention. Even though these techniques compress the spectral dimension and reduce spectral redundancy, they frequently introduce noise due to illumination and imaging equipment (Prasad and Bruce 2008, Bandos *et al*. 2009, Uddin *et al*. 2022). PCA captures essential dataset variations, reducing dimensionality while preserving key characteristics. LDA emphasises class separation, making it ideal for enhancing classification. ICA and MNF address signal-to-noise ratio problems, with ICA isolating independent sources of variation and MNF prioritising information-rich components while reducing noise influence. These approaches collectively offer strategies to overcome the challenges of hyperspectral data classification, balancing dimensionality reduction with feature selection and noise mitigation for improved accuracy and reliability.

1.2. DL-based approach

In recent years, DL revolutionised computer vision by automating complex feature extraction, achieving success in image classification (Li *et al*. 2018, Islam *et al*. 2023), target recognition (Zhang *et al*. 2015), and semantic segmentation (Pierdicca *et al*. 2020). Researchers have begun applying DL to HSI classification, starting with (Yuan *et al*. 2018, Lei *et al*. 2022, Hong *et al*. 2021, Sen and Keles 2022, Clark *et al*. 2023), address HSI-specific challenges. However, a standard limitation of some early DL methods for HSI is their exclusive focus on spectral information, explicitly relying on the spectral vector of the pixel for classification, which neglects the valuable spatial context of the images (Chen *et al*. 2015, Zhang *et al*. 2015, Yuan *et al*. 2018). This oversight results in the loss of crucial spatial details and hampers the classification's performance. Besides this, researchers have begun emphasising spatial structure's significance in HSIs. Incorporating spatial information, some use two-dimensional convolutional neural networks (2D CNNs) (Fang *et al*. 2015, Makantasis *et al*. 2015). Innovative architectures, like Makantasis *et al*. (2015), pack pixels into cubes to capture spatial context, improving spatial information extraction. Other methods, such as Li *et al*. (2017), exploit pixel similarity for labelling. Small-scale

data-driven techniques, like the multi-grained network by Pan *et al*. (2018), tuna swarm optimisation based DenseU-Net by Balaji *et al*. (2023), and a multiview calibrated prototype-learning framework designed for few-shot HSI classification introduced by Yu *et al*. (2022) to address limited HSI samples. Chen *et al*. (2023) introduces a multiscale superpixelwise random patched fusion network and adaptive multi-feature fusion graph convolution network by Liu *et al*. (2023) designed to enhance HSI classification.

CNNs exhibit strong feature extraction capabilities, allowing them to learn high-level abstract features through module stacking and increased network depth. However, deeper networks introduce additional parameters and longer training times. To alleviate these issues, GPUs have been developed to reduce training time for large parameter networks (Nickolls and Dally 2010). Another concern is gradient vanishing when the gradients of the loss function become extremely small during back-propagation through many layers, which can degrade classification performance by limiting the network's ability to learn complex patterns. To address this, He *et al*. (2016) proposed residual networks with skip connections to ensure smooth gradient circulation. Residual networks have gained popularity in computer vision and have been applied to HSI classification. For example, Zhong *et al*. (2018) designed an SSRN with residual blocks for discriminative feature learning, even with limited training samples. HSI classification techniques based on CNN mostly use 2D CNN and 3D CNN. In Ben Hamida *et al*. (2018), a 3D CNN-based model utilising different sizes of 3D convolutions is proposed to process spatial and spectral components simultaneously, achieving a reduced parameter model. He *et al*. (2019) presents a handcrafted feature extraction method based on multiscale covariance maps, providing improved robustness and classification performance. Fast and compact 3D-CNN (Fast 3D-CNN) employs incremental PCA (iPCA) and 3D-2D CNN for spectral-spatial feature reduction and classification of HSIs (Ahmad *et al*. 2020). Spectral Network (SpectralNET), introducing a wavelet-based architecture to extract both spatial and spectral information for HSI classification (Chakraborty and Trehan 2021). Hybrid Spectral Network (HybridSN) employed three 3D CNNs and a 2D CNN in their network architecture (Roy *et al*. 2020). Guo and Zhu (2019), introduced an HSI classification algorithm that combines spectral and spatial features through network transformations, proving its effectiveness. Similarly, Li *et al*. (2020) improved feature extraction by using a 2D convolutional network with separate branches for spatial and spectral features, leading to better classification performance. Finally, Ge *et al*. (2020) combines the strengths of 2D and 3D CNNs with multi-branch feature fusion. This approach enhances the feature extraction capability by integrating spectral and spatial information more effectively, leading to improved accuracy compared to traditional CNN-based approaches.

Among DL models, CNNs excel in extracting spectral and spatial features for HSI classification. However, challenges include convergence to local minima, information loss in pooling layers, and hyperparameter selection for single-branch networks. Utilizing 2D CNN alone may miss interband information, while 3D CNN captures joint spectral-spatial features but might underperform when textures are similar across bands. Combining 3D and 2D CNNs with different kernel sizes enhances spatial representation. A multi-branch network that extracts HSI features further enriches extracted features by fusing them. To address these challenges, we propose an HSI classification method based on 2D-3D CNNs and multi-branch feature fusion. Traditional PCA assumes a single global model for an entire HSI, which may not

accurately capture its complex spatial and spectral variations. As an alternative, Segmented PCA divides the image into smaller regions and applies PCA separately to each segment, offering a more localised and adaptive representation preserving spatial and spectral information. In our approach, we segment the HSI datacube into L spectral regions, applying classical PCA to each. We then employ a neural network with L branches, combining 3D CNN and 2D CNN, to extract features from the segmented principal components. These features are integrated with the spectral dimension and processed through fully connected layers for final classification.

The key contributions of this paper are summarised as follows:

(1) Our innovative hybrid approach integrates Segmented PCA with a 3D-2D CNN, augmented by multibranch feature fusion, advancing both theory and technique. This unified framework optimally extracts spectral-spatial features, contributing to a more robust model that significantly enhances classification accuracy.
(2) Beyond theory, our method introduces a unique strategy by incorporating data from diverse sub-regions within HSI cubes. This practical innovation results in a more comprehensive representation of the data, improving classification performance in real-world scenarios.
(3) We perform rigorous evaluations across three benchmark datasets to provide empirical evidence, validating the effectiveness of our approach and reinforcing its broad applicability.

The rest of the sections of this article are structured as follows. Section 2 presents an in-depth discussion on the theoretical basis related to the topic. The proposed neural network approach and its particular implementation steps are outlined in Section 3. The experimental results obtained and their subsequent analysis are thoroughly discussed in Section 4. Finally, the conclusions drawn from this study are summarised in Section 5.

2. Background theory

2.1. Principal component analysis (PCA)

PCA is a popular feature extraction and reduction technique for analysing HSI data. It attempts to maintain as much information as possible while transforming the high-dimensional HSI data into a space with fewer dimensions. PCA achieves this by identifying the directions, known as PCs, that capture the maximum variance in the data (Rodarmel and Shan 2002). These PCs are sorted according to how much they contribute to the overall variance and are orthogonal to one another. By selecting a subset of the PCs, one can effectively reduce the features of the HSI data, facilitating visualisation, denoising, and feature extraction for improved analysis and classification in HSI applications. For PCA implementation: let x_n represent the spectral vector of a sample pixel in the HSI data matrix, denoted as $x_n = [\mathrm{x}_{n1}\mathrm{x}_{n2}\ldots\mathrm{x}_{nF}]^T$, where $n \in [1, S]$ and $S = X \times Y$, with X and Y is the spatial dimensions of the data matrix $\mathbf{D}$ of size $F \times S$. The zero-mean image $\mathbf{I} = [\mathbf{I}_1\mathbf{I}_2\ldots\mathbf{I}_n]$ is derived from $\mathbf{D}$, where $\mathbf{I}_n = \boldsymbol{x}_n - \mathbf{M} = [\mathbf{I}_{n1}\mathbf{I}_{n2}\ldots\mathbf{I}_{nF}]^T$, and $\mathbf{M}$ is defined as $\mathbf{M} = \frac{1}{S}\sum n = 1^S \boldsymbol{x}_n$. Subsequently,

the covariance matrix $\mathbf{C} = \frac{1}{S}\mathbf{I}\mathbf{I}^T$ is computed using the following Eigendecomposition equation:

$$\mathbf{C} = \mathbf{V}\mathbf{E}\mathbf{V}^T, \tag{1}$$

where E and V denote the matrices of corresponding eigenvalues and eigenvectors (also known as PCs), respectively. Typically, q eigenvectors are selected to form $\mathbf{W}$, an $F \times q$ dimensional matrix, where $q \leq F$ and often $q \ll F$. The conventional approach for choosing the top q PCs is based on ranking the eigenvalues in descending order. Finally, the projection matrix $\mathbf{Z}$ of $\mathbf{D}$ is obtained as $\mathbf{Z} = \mathbf{W}^T \times \mathbf{I}$.

2.2. Convolutional neural network (CNN)

CNN is a type of feedforward neural network known for its excellent performance in large-scale image processing (Aloysius and Geetha 2017, Li *et al*. 2019, 2022, Liu *et al*. 2021, Xie *et al*. 2022, Zhao *et al*. 2023). It operates by having artificial neurons that respond to specific portions of the surrounding cells within the coverage area. CNN incorporates two fundamental design principles. First, it utilises the 2D structure of an image, recognising that pixels in adjacent regions are typically highly correlated. Second, CNN relies on feature sharing, where each channel (output feature map) is generated through convolution using the same filter across all locations. The CNN architecture consists of four main components: the convolutional layer, activation function, pooling layer, and fully connected layer. The convolutional layer performs calculations according to the following equation:

$$x_i^l = \sum_{j=1}^{f} \phi(w_{ji}^l * x_j^{l-1} + b_i^l), \tag{2}$$

where, x_j^{l-1} represents the jth feature map of the $(l-1)$th layer, x_i^l represents the ith feature map of the lth layer, and f denotes the number of input feature maps. The parameters w_{ji}^l and b_i^l correspond to the weight and bias, respectively, which are randomly initialised. The function $\phi(.)$ denotes a nonlinear activation function, with the rectified linear unit (ReLU) function being employed in this article. The symbol $*$ denotes the convolution operation. Following the convolutional layer, the pooling layer is typically employed to reduce the spatial dimension of the data, effectively reducing network parameters and preventing overfitting. Finally, the fully connected layer connects multiple neurons to the previous layer and transmits output values to the classifier. The training of all neural network parameters utilises the backpropagation algorithm.

The flexible combination of the convolutional layer, pooling layer, and fully connected layer via CNN enables the network to precisely obtain information from images. HSI classification techniques based on CNN mostly use 2D CNN and 3D CNN.

While combining 2D CNN with 3D CNN can somewhat enhance performance, determining the optimal number of network layers is challenging. Neither excessively deep nor shallow networks can effectively extract image features. In the case of multibranch feature extraction, features from different branches are fused together to make combined features. Feature fusion methods can be categorised into early and late fusions based on the fusion order and prediction. Early fusion methods involve initially merging multilayer

features and training a predictor using the fused features. This approach is sometimes called skip connection and employs concatenation or addition operations. In contrast, late fusion methods enhance performance by fusing the detection results obtained from different layers. In this article, we employ concatenation for feature fusion.

3. Proposed approach

The proposed approach for HSI classification, depicted in Figure 1, utilises a combination of Seg-PCA, 3D-2D CNN, and multi-branch feature fusion to enhance the accuracy of classification. Initially, the Seg-PCA technique is applied to reduce the dimensionality of the original dataset. Subsequently, the reduced data is fed into a multi-branch network, where features are extracted for L different segments' PCA data using the powerful 3D-2D CNN architecture. These extracted features are then fused using the concatenation operation, allowing for the combination of information from various segments. Finally, the fused features are passed through a series of fully connected (dense, dropout) layers, followed by a softmax layer for the ultimate classification of the HSI data. By incorporating Seg-PCA, 3D-2D CNN, and multi-branch feature fusion, the proposed model aims to achieve improved performance in HSI classification tasks. The details of the proposed approach are presented in algorithm 1, and each method inside the proposed approach is described with individual subsections.

3.1. Seg-PCA

Previous studies have shown that PCA can extract more significant features when the original HSI bands exhibit a high correlation, resulting in a large covariance matrix. However, this approach incurs a higher computational cost. Since PCA considers global statistics, it may overlook subtle information in the HSI that could be valuable for accurate categorisation. To improve performance, Seg-PCA has been proposed and applied in different research as an alternative to traditional PCA, specifically tailored for handling highly correlated blocks with low inter-correlations (Jia and Richards 1999, Tsai *et al*. 2007, Uddin *et al*. 2021a, 2021b). We first apply Seg-PCA to

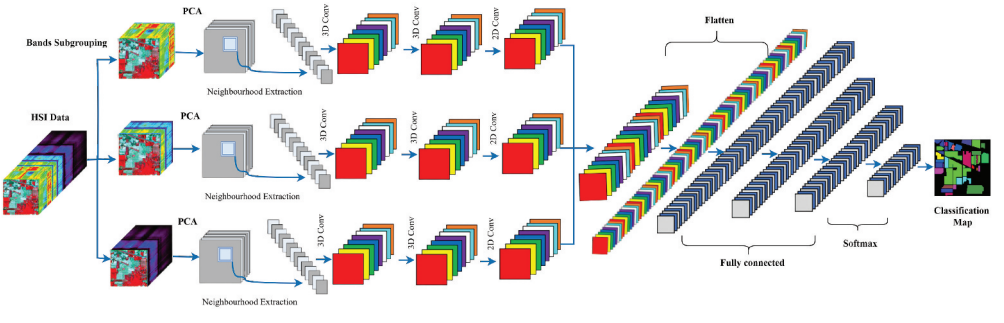


Figure 1. Overall flow diagram of the proposed Seg-PCA and 3D-2D CNN method for the HSI classification leveraging multi-branch fused features.

HSI to remove the spectral redundancy at the global and local statistics levels. In Seg-PCA implementation, the HSI data matrix is initially divided into L subgroup datasets based on band correlations (Uddin *et al*. 2021a). In this proposed method implementation, the value of L ranges from 2 to 4. Typically, strongly correlated bands are grouped in the same subgroup to capture the local characteristics of the dataset. For, Seg-PCA implementation: the sub-grouped data matrices are denoted as $\mathbf{D}_t$, where $t \in [1, 2, 3]$, and for each segmented data matrix, n_t represents the number of consecutive bands. For example, $\mathbf{D}_1 = [\mathbf{I}_1^{'}\mathbf{I}_2^{'} \ \ldots \ \mathbf{I}_{n_1}^{'}]$, $\mathbf{D}_2 = [\mathbf{I}_1^{''}\mathbf{I}_2^{''} \ \ldots \ \mathbf{I}_{n_2}^{''}]$, and so on. Here, $\mathbf{I}_j^{'}$ consists of the first n_1 rows of the corresponding $\mathbf{I}_j$, while $\mathbf{I}_j^{''}$ includes the first n_2 rows (excluding the initial n_1 rows) of $\mathbf{I}_j$. These operations are performed for each $j, n \in [1, S]$. Subsequently, the Eigendecomposition operation is applied to each $\mathbf{D}_t$ to calculate the covariance matrix. The overall projection matrix is obtained by sequentially combining the projection matrices of each subgroup's $\mathbf{D}_t$.

The Seg-PCA technique selectively reduces the number of spectral bands from F to B_1, B_2, and B_3, representing the extracted and selected PC dimensions for three distinct subgroups when L is 3. However, the spatial dimensions of the data matrix remain unchanged, denoted as $S = X \times Y$, where $\mathbf{D} = X \times Y \times F$ represents the overall input data cube. The modified input data after Seg-PCA is represented by $\mathbf{Z}_t = X \times Y \times B_t$, where $\mathbf{Z}_t$ corresponds to the modified input data cube after Seg-PCA and t takes the values 1, 2, and 3 to indicate the segment number.

3.2. Neighbourhood extraction process

The HSI data cube is split into tiny overlapping 3D patches to execute our multi-branch 3D-2D CNN model. These patches, denoted as $P_t = s \times s \times B_t$, are created from the Seg-PCA reduced data cube $\mathbf{Z}_t$ and are centered at spatial location (i, j). Each patch covers a spatial extent of $s \times s$ and contains all B_t spectral bands' PCs. The total of the generated 3D patches, denoted as n, from $\mathbf{Z}t$ is determined by $(X - s + 1) \times (Y - s + 1)$, where X and Y represent the spatial dimensions of the data cube. Consequently, the 3D patch located at (i, j), represented as Pt, i, j, encompasses the width from $i - (s - 1)/2$ to $i + (s - 1)/2$, height from $j - (s - 1)/2$ to $j + (s - 1)/2$, and includes all B_t spectral bands of the Seg-PCA reduced data cube $\mathbf{Z}_t$ for each subgroup t.

Algorithm 1 Procedure of the proposed HSI classification

Input: Data matrix **D**, Sub-block number L, Number of PCs B, Window size $s \times s$.
Output: The final classification results and classification map.

1: **Begin: D** take the input data matrix of size $S \times F$ where F is the number of spectral bands and S is the number of spatial pixels.
2: **Divide D** into L sub-group based on the average correlation.
3: **for** $t = 1$ to L **do**
4: **Apply** classical PCA to reduce the spectral dimensionality of subgroup data D_t to B.
5: **Perform** mean adjustment on PCA transformed data.
6: **Perform neighborhood extraction** to generate tiny 3D overlapping patches $P_t = s \times s \times B$. {*Multibranch 3D-2D CNN start*.}
7: **Apply 3D CNN** with kernel size $3 \times 3 \times 3$ and is composed of a total of 6 kernels by equation 3.

(*Continued*)

Algorithm 1 Procedure of the proposed HSI classification

8: **Apply 3D CNN** with kernel size $3 \times 3 \times 3$ and is composed of a total of 12 kernels by equation 3.
9: **Reshape** the data for 2D CNN.
10: **Apply 2D CNN** with kernel size 3×3 and is composed of a total of 64 kernels by equation 4. {*Multibranch 3D-2D CNN end*.}
11: **end for**
12: **Concatenate** the extracted features for all t after 2D CNN. {*feature fusion*}
13: **Flatten** the outputs of concatenate layers.
14: **Perform the dense and dropout** three times with each 40% drop and ReLU activation function.
15: **Compute output** for the last dense layer with a softmax activation function.
16: **Train** the overall model designed from step 7 to step 15 by optimizer:adam, loss: categorical crossentropy, learning rate:0.001, the decay rate:0.000001, the batch size:256, and epochs:100. {*All of the parameters are set by trial and error based on classification results*.}
17: **Test** the model and calculate the results.
18: **Generate** the classification map for the full dataset.

3.3. Multi-branch 3D-2D CNN feature fusion

In our proposed methodology, we introduce a novel and sophisticated multi-branch 3D-2D CNN architecture designed for enhanced feature extraction in HSI classification. Specifically, when the parameter L is set to 3, our architecture incorporates three distinct branches. The input for three branches is from three subgroups of the PCA transformed and reduced dataset. Incorporating data from diverse sub-regions within HSI cubes. This practical innovation results in a more comprehensive representation of the data, boosting classification performance. Each branch comprises two 3D convolution layers and one 2D convolution layer strategically chosen to capture intricate spectral-spatial relationships within the input data. The crucial step of feature fusion follows the convolutional operations. We seamlessly concatenate the output from the three branches, creating a fused representation that integrates diverse spectral and spatial features. This fusion process is instrumental in achieving a more comprehensive and robust model, enriching the information available for subsequent layers. To facilitate downstream processing, we employ a flattened layer that transforms the multidimensional feature maps into a single, long, continuous linear vector. This vectorised representation serves as the input for three fully connected layers and an output layer, collectively forming the latter part of our architecture. This intricate design allows the model to discern complex patterns and relationships within the hyperspectral data, contributing to improved classification accuracy. The core of our approach lies in the 3D convolution operation, where a three-dimensional kernel convolves with the input data. This operation effectively captures spectral information in multiple contiguous bands of the hyperspectral input layer. By doing so, our model excels in leveraging the rich spectral content inherent in hyperspectral images, enhancing its ability to discriminate between different classes. In the 3D convolution equation, the activation value at a spatial position (i, j, k) in the f th feature map of the l th layer, denoted as $x_{l,f}^{i,j,k}$, is computed by applying the activation function ϕ to the sum of the bias parameter $b_{l,f}$ and the weighted sum of the input values from the previous layer using the kernel weights $w_{l,f,a}$. The kernel has a width of $2\xi + 1$, a height of $2\delta + 1$, and a depth of $2\eta + 1$ along with the spectral dimension, where ξ, δ, and η determine the kernel size. The weight parameter $w_{l,f,a}$ represents the weight value for the f th feature map of the l th

layer and the α th spectral band. The computation of the activation function is shown below:

$$x_{l,f}^{i,j,k} = \phi\left(b_{l,f} + \sum_{\alpha=1}^{\lambda_{l-1}} \sum_{\beta=-\eta}^{\eta} \sum_{\gamma=-\xi}^{\xi} \sum_{\sigma=-\delta}^{\delta} w_{l,f,\alpha}^{\sigma,\gamma,\beta} * x_{l-1,\alpha}^{i+\sigma,j+\gamma,k+\beta}\right). \tag{3}$$

In the case of 2D-CNN, the input data are convolved with 2D kernels. The convolution operation involves computing the sum of the dot product between the input data and the kernel, which is strided over the input data to cover the full spatial dimension. The resulting convolved features are then passed through the activation function to introduce nonlinearity in the model. In the equation for 2D convolution, the activation value at spatial position (i, j) in the fth feature map of the lth layer, denoted as $x_{l,f}^{i,j}$, is computed by applying the activation function ϕ to the sum of the bias parameter $b_{l,f}$ and the weighted sum of the input values from the previous layer using the kernel weights $w_{l,f,\alpha}$ as follow:

$$x_{l,f}^{i,j} = \phi\left(b_{l,f} + \sum_{\alpha=1}^{\lambda_{l-1}} \sum_{\gamma=-\xi}^{\xi} \sum_{\sigma=-\delta}^{\delta} w_{l,f,\alpha}^{\sigma,\gamma} * x_{l-1,\alpha}^{i+\sigma,j+\gamma,}\right). \tag{4}$$

The kernel has a width of $2\xi + 1$ and a height of $2\delta + 1$, and it does not include the spectral dimension as in the 3D convolution equation.

The parameters of the CNN, such as bias and kernel weights, are typically trained using supervised approaches with gradient descent optimisation. Conventional 2D CNNs apply convolutions over spatial dimensions only, which limits their ability to capture spectral information. In contrast, 3D CNNs can extract both spatial and spectral features simultaneously but with higher computational complexity. To leverage the feature learning capabilities of both 3D and 2D CNNs, our proposed multi-branch model combines them to extract different features and fuse them for a more discriminative representation. The flattening layer is applied to preserve spatial information while reducing spectral dimensionality.

3.4. Model architecture and parameter details

The model's architecture and parameter details are summarised in Table 1, with the highest number of parameters found in the first dense layer. The last dense layer's number of nodes corresponds to the number of classes in the dataset. As shown in Table 1, each branch of the proposed model consists of convolutional layers with different kernel sizes and numbers of kernels. The first 3D CNN in each branch has a kernel size of $3 \times 3 \times 3$ and is composed of 6 kernels. The second 3D CNN has the same kernel size of $3 \times 3 \times 3$ but contains 12 kernels. Finally, the 2D CNN in each branch has a kernel size of 3×3 and includes 64 kernels. These convolutional layers with varying kernel sizes and numbers contribute to the extraction of different spatial and spectral features from the HSI data, enhancing the discriminative power of the model. However, the proposed model's total number of trainable weight parameters is 35,689,076 for the Indian Pines (IP) dataset. These weights are randomly initialised and trained using the back-propagation algorithm with the Adam optimiser and softmax loss. The network is trained with mini-batches of size 256 for 100 epochs without batch normalisation or data augmentation. ReLU activation function

Table 1. Layer-wise parameter details for the proposed model architecture with 25 × 25 window size based on IP dataset.

Layer Type	Output Shape	Number of Parameter	Repeat Times
input (InputLayer)	25, 25, 5, 1	0	3
conv3d-1 (Conv3D)	23, 23, 3, 6	168	3
conv3d-2 (Conv3D)	21, 21, 1, 12	1956	3
reshape (Reshape)	21, 21, 12	0	3
conv2d (Conv2D)	19, 19, 64	6976	3
concatenate (Concatenate)	19, 19, 192	0	
flatten (Flatten)	69312	0	
dense-1 (Dense)	512	35488256	
dropout-1 (Dropout)	512	0	
dense-2 (Dense)	256	131328	
dropout-2 (Dropout)	256	0	
dense-3 (Dense)	128	32896	
dropout-3 (Dropout)	128	0	
output (Output)	16	2064	

was employed throughout the network, except for the output layer, as it is a commonly used activation function in HSI classification. A softmax activation function generates class probabilities for multi-class classification tasks in the output layer.

4. Experiment and result analysis

4.1. Dataset description

We conducted experiments using three publicly available HSI datasets: Indian Pines (IP), University of Pavia (PU), and Salinas Scene (SA). The IP dataset consists of images with a spatial dimension of 145 × 145 pixels and 224 spectral bands in the wavelength range of 400 to 2500 *nm*. We excluded 24 spectral bands corresponding to water absorption regions. The ground truth of the IP dataset provides labels for 16 vegetation classes (Baumgardner *et al*. 2015). The UP dataset comprises images with a spatial dimension of 610 × 340 pixels and 103 spectral bands in the wavelength range of 430 to 860 *nm*. The ground truth of the UP dataset classifies urban land cover into nine classes (Gamba 2004). The SA dataset includes images with a spatial dimension of 512 × 217 pixels and 224 spectral bands in the wavelength range of 360 to 2500 *nm*. We removed 20 spectral bands affected by water absorption. The SA dataset consists of 16 different classes (GIC 2021). These datasets were chosen for their diversity in spatial dimensions, spectral bands, and land-cover classes, allowing us to evaluate the performance and generalisation capability of our proposed method across different HSI scenarios.

4.2. Experiment design and parameter setup

We divided the labelled samples into training validation and test subsets using different ratios for each of the three datasets. The ratios used were 5%–95%, 10%–90%, 15%–85%, and 20%–80%. For example, in the case of a 10%-90% ratio, 10% of the labelled sampled was randomly assigned to training and validation purposes, and the remaining 90% was assigned to test the model. During the CNN training process, 90% of the training and validation labelled samples were used to determine the weights and biases of each

neuron, and the remaining 10% were utilised to determine overfitting and inform network design choices.

In our proposed model, first, we apply Seg-PCA. For, Seg-PCA, we segment' each HSI dataset into L segments by analyzing the average correlation that exists in the segmented subgroups. Here, L ranges from 2 to 4. Then, we select the optimum segmentation based on their performances. After Seg-PCA, we scale the data by the min-max scaler of the Python sklearn preprocessing library. We set the learning rate to 0.001, the decay rate to 0.000001, the batch size to 256, and epochs to 100 based on the classification results. We extracted 3D patches from the input volume with a spatial dimension of 25×25 and $F_{spectral}$ spectral features to ensure a fair comparison. $F_{spectral}$ represents the number of input spectral features. In our experiment, we applied Seg-PCA for spectral feature reduction and selected the top-ranked 5 (five) features from each segment based on their cumulative explained variance. Cumulative variances are increased by taking more PCs. Figure 2 shows the cumulative explained variance versus the number of PCs relationship for IP data. From Figure 2 we can observe that the top-ranked 5 features do not always carry the maximum variances (near about 100%). Thus, we perform another experiment with the 10 (ten) spectral features. Which is the near-optimal number of PCs to input into the 3D-2D CNN branches from each segmented PCA.

To evaluate the classification performance, we used three commonly used quantitative metrics: (i) Overall Accuracy (OA): The ratio of correctly classified samples to the total number of test samples, (ii) Average Accuracy (AA): The average of classwise classification accuracies, and (iii) Kappa Coefficient (Kappa): A statistical measure of agreement among qualitative items. Additionally, we analysed the training loss and training accuracy during the training process. Finally, we produce the classification maps for more visualisation of the results and to further validate the effectiveness of the proposed method.

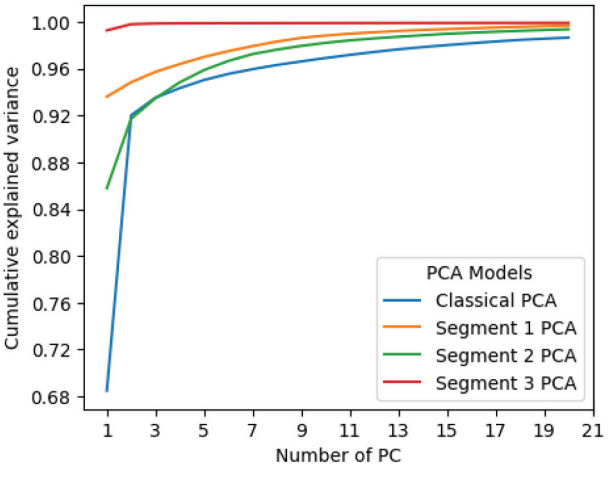


Figure 2. Cumulative explained variance vs. number of PC for IP dataset for different PCA models, including classical PCA on the entire dataset and PCA on the segmented datasets.

4.3. Impact of dataset segmentation

Initially, we divide each dataset into L segments by analyzing their average band-to-band correlation value in the correlation matrix (Jia and Richards 1999, Uddin *et al*. 2021a, Afjal *et al*. 2022). We perform three distinct segmentations for the IP and SA datasets, Seg-2, Seg-3, and Seg-4, where L is 2, 3, and 4, respectively. Similarly, we divide the PU dataset into two distinct segments namely, Seg-2 and Seg-3, where L is 2 and 3 respectively. The image representation of band-to-band correlation matrices for various HSI datasets is illustrated in Figure 3. Specifically, Figure 3(a) displays the band-to-band correlation matrix for the IP, (b) PU, and (d) SA HSI. In the context of Seg-PCA, we exclusively considered correlation values whose absolute magnitudes surpassed a predetermined threshold. To facilitate correlation-based segmentation, we set the threshold at 0.5. This threshold was determined by visually inspecting the band-to-band correlation matrix image, where we identified edges in the matrix image, as presented in Figure 3(d). An exemplar segmentation result is depicted in image form in Figure 3(d) for the SA HSI

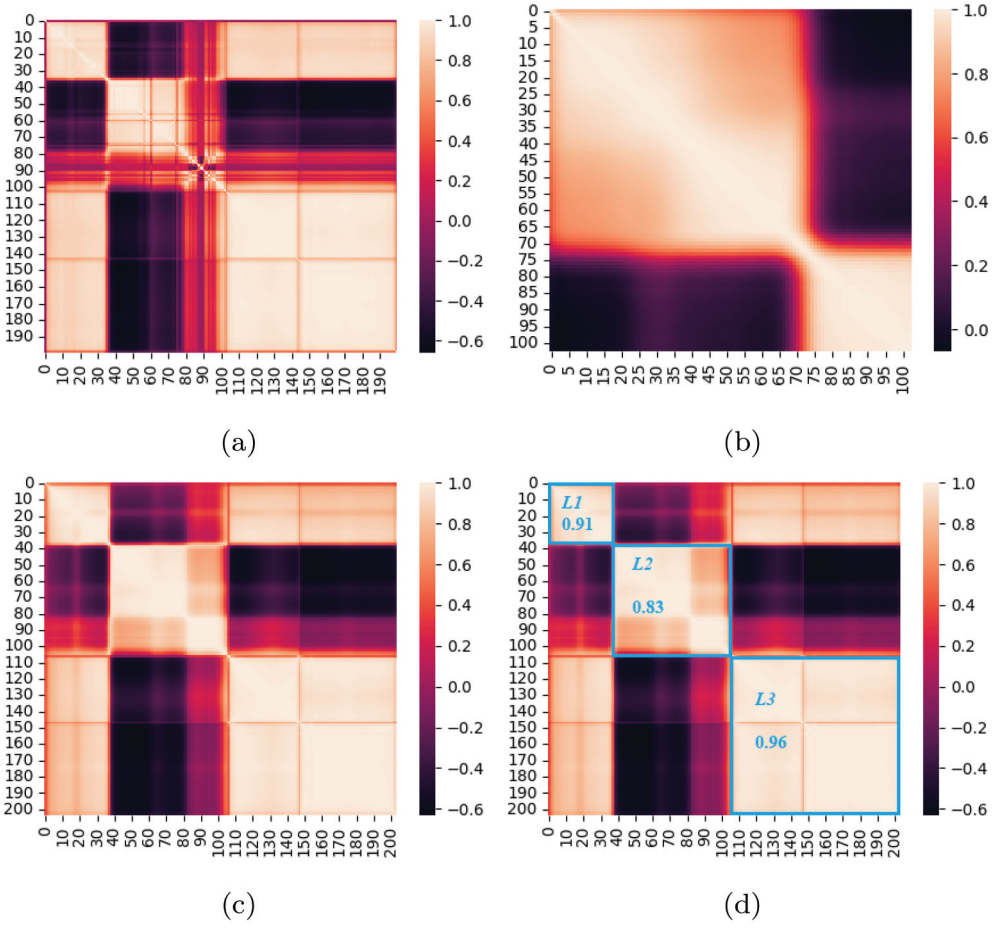


Figure 3. Image representation of band-to-band correlation matrix of (a) IP, (b) PU, (c) SA, and (d) segmentation on correlation matrix image of SA when L equals 3.

Table 2. Details of datasets segmentation for Seg-PCA.

		Number of segmented datasets							
		IP			SA			PU	
L No.	Parameter	Seg-2	Seg-3	Seg-4	Seg-2	Seg-3	Seg-4	Seg-2	Seg-3
1	Range of bands	1–34	1–34	1–34	1–36	1–36	1–36	1–37	1–37
	Number of bands	34	34	34	36	36	36	37	37
	Average correlation	0.88	0.88	0.88	0.91	0.91	0.91	0.96	0.96
2	Range of bands	35–200	35–102	35–80	37–204	37–106	37–81	38–103	38–72
	Number of bands	166	68	46	168	70	45	66	35
	Average correlation	0.35	0.57	0.83	0.34	0.83	0.90	0.58	0.96
3	Range of bands	N/A	103–200	81–102	N/A	107–204	82–106	N/A	73–103
	Number of bands		98	22		98	25		31
	Average correlation		0.95	0.52		0.96	0.94		0.95
4	Range of bands	N/A	N/A	103–200	N/A	N/A	107–204	N/A	N/A
	Number of bands			98			98		
	Average correlation			0.95			0.96		

L No. denote the segment number.

Table 3. Classification results for different segmentation when the number of spectral features is 5 and 10% training data are used.

	IP			SA			PU		
Number of segments	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
2 (Seg-2)	94.76	81.39	94.02	99.96	99.96	99.96	98.67	97.01	98.23
3 (Seg-3)	97.05	82.63	96.52	99.99	99.97	99.98	99.78	99.31	99.63
4 (Seg-4)	97.07	82.57	96.49	99.98	99.96	99.97		N/A	

dataset. This segmentation process resulted in three primary subgroups of bands. The details of dataset segmentations are shown in Table 2. To select the optimum segmentation, we experiment with all the segmentation individually for each of the three datasets. In this experiment, for fare comparison, we select the number of spectral features as 5 and 10% training data to train and validate the proposed model. Classification accuracy results are shown in Table 3 for different segmentation. Results indicate that the Seg-3 or three-segment dataset performs better than Seg-2 and Seg-4 except for IP Seg-4 results. Seg-4 needs more computation than Seg-3 because it uses four 3D-2D CNN branches instead of three. Thus, we take *L* equal to 3 and divide the entire dataset into three subgroups.

4.4. Classification results

The classification results were evaluated using OA, AA, and Kappa as described in the experimental design. The accuracies were measured in percentages for both the proposed method and state-of-the-art methods, including SVM on original data, Segmented PCA + SVM, 2D-CNN, Fast 3D-CNN, SpectralNet, HybridSN, and 2D-3D multibranch fusion. These state-of-the-art experiments are described as follows:

(1) 2D-CNN (Makantasis *et al.* 2015): Apply randomised PCA for dimensionality reduction. The network consists of two convolutional layers with dimensions 3×3 and filter counts of 30 and 90, respectively. Additionally, there are two fully connected layers, each incorporating dropout rates of 25% and 50%, respectively, and an

output layer. The activation function employed is the ReLU, and the classifier utilises the SoftMax function.

(2) Fast 3D-CNN (Ahmad *et al*. 2020): Utilize incremental PCA for dimensionality reduction. The network consists of four convolutional layers with dimensions $3 \times 3 \times 7$, $3 \times 3 \times 5$, $3 \times 3 \times 3$, and $3 \times 3 \times 3$, respectively, with filter counts of 8, 16, 32, and 64. Additionally, there are three fully connected layers.

(3) SpectralNet (Chakraborty and Trehan 2021): Apply factor analysis for dimensionality reduction. Subsequently, execute a three-level Wavelet CNN decomposition. A Wavelet CNN employs layers of wavelet transforms to accentuate spectral features.

(4) HybridSN (Roy *et al*. 2020): Apply PCA for dimensionality reduction. This architecture comprises three layers of 3D convolution, one layer of 2D convolution, and three fully connected layers.

(5) 2D-3D multibranch (Ge *et al*. 2020): Utilize PCA for dimensionality reduction. This architecture consists of three distinct branches, all sharing PCA-extracted same features. The first branch includes one 3D and one 2D layer, the second incorporates two 3D and one 2D layer, and the third employs three 3D and one 2D layer. Subsequently, the features from these different branches are fused and passed through four fully connected layers, culminating in an output layer.

Table 4 presents the classification results for five (5) spectral features and four different training-testing separations. The OA of the best-performing methods under different conditions is highlighted in bold. From the table, it can be observed that the proposed method consistently achieves better results compared to the other methods in most cases. Among the three datasets, the PU dataset has the lowest number of categories, consisting of only nine classes. This makes it relatively easier to classify compared to the other datasets. On the other hand, the SA dataset has a larger spatial size and the maximum number of spectral bands, which provides more discriminative information for classification. Therefore, the listed methods generally achieve higher classification performance on the SA dataset. The IP dataset has a relatively small spatial size but contains 16 categories, which poses a challenge for accurate classification and results in lower classification accuracy compared to the other datasets. The traditional SVM applied directly to the original data yields poor classification performance. However, when the data is preprocessed using Seg-PCA and combined with SVM, there is a significant improvement in classification accuracy.

The 2D-CNN and Fast 3D-CNN methods exhibit lower classification performances, likely due to the limited utilisation of spectral information. SpectralNet, on the other hand, achieves moderate classification performances by incorporating spectral and spatial features. HybridSN, which combines several 3D CNNs with a 2D CNN, proves to be a simple and effective approach, yielding discriminative features. The 2D-3D Multibranch method, with its different kernel design and deeper level multibranch feature fusion, effectively prevents overfitting and makes optimal use of the joint spectral-spatial features of HSI, resulting in good classification performance. Indeed, the proposed method takes advantage of both spatial and spectral features of HSI data. Additionally, by extracting different image features from segmented subgroups' PCs and fusing them together, the network learns representative and discriminative features. This feature

Table 4. Classification results for the proposed and state-of-the-art methods when the number of spectral feature is 5.

Method	Training Samples	IP			SA			PU		
		OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
Original data+SVM	5%	41.92	22.37	29.01	58.41	44.77	52.67	60.02	36.07	41.32
Chang and Lin (2011)	10%	42.50	23.09	29.92	58.87	46.71	53.32	60.21	36.13	41.58
	15%	42.30	22.34	29.39	59.05	47.19	53.56	60.24	36.15	41.65
	20%	42.43	22.26	29.31	59.17	47.57	53.70	60.32	36.26	41.86
SegPCA+SVM	5%	64.57	53.45	58.94	91.56	94.98	90.59	88.35	79.00	84.23
	10%	68.64	59.08	63.60	91.94	95.48	91.00	91.05	85.63	87.99
	15%	71.68	62.83	67.21	92.25	95.78	91.36	91.79	87.40	89.00
	20%	73.54	64.79	69.39	92.40	96.01	91.51	92.22	88.29	89.59
2D-CNN	5%	19.68	29.63	12.49	75.47	88.00	73.10	73.47	82.10	66.91
Makantasis *et al*. (2015)	10%	41.33	53.79	35.63	80.01	93.66	78.10	89.70	88.80	86.46
	15%	58.34	68.45	53.03	83.82	94.79	82.21	91.86	92.71	89.36
	20%	55.22	70.65	50.25	85.64	95.51	84.19	94.45	94.67	92.70
Fast 3D-CNN	5%	52.96	35.77	44.94	95.71	95.57	95.23	84.46	72.55	78.82
Ahmad *et al*. (2020)	10%	50.27	31.19	41.27	80.17	82.14	82.22	91.06	81.95	88.06
	15%	55.56	42.78	49.02	91.89	89.59	90.96	95.99	90.81	94.67
	20%	69.52	44.06	64.53	97.97	97.59	97.73	96.40	92.25	95.22
SpectralNet	5%	27.66	13.26	18.40	99.70	99.86	99.66	98.04	96.86	97.41
Chakraborty and Trehan (2021)	10%	94.24	74.39	93.43	99.43	99.72	99.36	99.15	99.33	98.87
	15%	97.92	97.24	97.63	99.87	99.88	99.85	99.61	99.46	99.52
	20%	98.70	98.53	98.54	99.88	99.89	99.87	99.63	99.65	99.52
HybridSN	5%	87.92	78.72	86.12	99.86	99.84	99.84	**99.00**	98.02	98.67
Roy *et al*. (2020)	10%	96.34	93.51	95.83	99.93	99.93	99.92	99.44	99.17	99.28
	15%	97.91	98.03	97.62	99.22	99.74	99.23	**99.90**	99.87	99.85
	20%	98.25	97.95	98.02	99.92	99.93	99.91	99.90	99.88	99.87
2D-3D Multibranch	5%	**89.86**	70.83	88.37	99.87	99.92	99.81	98.74	96.95	98.34
Ge *et al*. (2020)	10%	96.42	86.55	95.85	99.91	99.88	99.84	96.25	94.51	94.97
	15%	97.51	91.84	97.12	**99.92**	99.94	99.87	99.85	99.86	99.81
	20%	98.65	97.78	98.42	99.93	99.94	99.88	99.89	99.91	99.86
Proposed Method	5%	89.69	69.78	87.25	**99.88**	99.87	99.85	98.86	98.56	98.39
	10%	**97.05**	82.63	96.52	**99.99**	99.97	99.98	**99.78**	99.31	99.63
	15%	**98.41**	87.46	97.44	99.77	99.66	99.62	99.87	99.80	99.79
	20%	**99.07**	98.94	98.46	**100.00**	99.99	100.00	**99.95**	99.88	99.89

fusion helps reduce intraclass differences and increase interclass differences, leading to improved classification performance.

Table 5 presents classification results using ten spectral features, following a similar setup to Table 4. Increasing the number of spectral features leads to improved performance across all investigated methods compared to the use of five spectral features in Table 4. In most cases, our proposed method outperforms state-of-the-art methods when considering both 5 and 10 spectral features as input. Class-wise classification accuracies, expressed as percentages, can be found in Tables 6–8 with 10 spectral features and 20% training samples for all three datasets. The results in Tables 5 and 6–8 demonstrate that the proposed methods achieve the highest OA of 99.27%, 100%, and 99.99% for IP, SA, and PU datasets, respectively, using 10 spectral features and 20% training samples. In comparison, HybridSN and the 2D-3D Multibranch model achieve OAs of 99.10% and 98.96% for IP, 99.99%, and 99.98% for SA, and 99.98% and 99.94% for PU datasets, respectively, under the same spectral features and training sample conditions. Table 4 showcases the superior performance of the proposed method when compared to the HybridSN and 2D-3D Multibranch models. Specifically, for the IP dataset with 5 spectral features and 20% training samples, the proposed method achieves an OA of 99.07%, while HybridSN and 2D-3D Multibranch models achieve OAs of 98.28% and 98.65%,

Table 5. Classification results for the proposed and state-of-the-art methods when the number of spectral feature is 10.

Method	Training Samples	IP			SA			PU		
		OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
Original data+SVM	5%	45.17	25.25	33.45	69.98	65.22	66.10	62.34	38.17	45.10
Chang and Lin (2011)	10%	46.11	26.17	34.72	70.93	67.13	67.22	62.49	38.43	45.27
	15%	46.43	26.34	35.21	71.47	68.07	67.81	62.63	38.46	45.52
	20%	46.44	26.89	35.41	71.74	68.46	68.13	62.62	38.53	45.56
SegPCA+SVM	5%	64.12	53.06	58.51	91.58	94.96	90.69	86.20	75.29	81.18
	10%	70.57	59.80	65.92	92.16	95.56	91.26	88.53	78.52	84.49
	15%	73.64	63.52	69.53	92.55	95.59	91.68	89.81	81.48	86.27
	20%	75.55	67.36	71.78	92.71	96.13	91.87	90.61	83.82	87.39
2D-CNN	5%	64.77	65.55	60.14	97.06	98.83	96.73	95.09	94.19	93.52
Makantasis *et al*. (2015)	10%	78.53	83.88	75.73	97.84	99.20	97.60	97.74	97.27	97.02
	15%	85.08	91.61	83.11	98.26	99.36	98.06	98.97	98.59	98.63
	20%	87.04	93.60	85.35	98.37	99.44	98.19	99.38	99.07	99.18
Fast 3D-CNN	5%	55.18	36.31	46.93	93.32	96.24	92.56	89.79	82.31	86.34
Ahmad *et al*. (2020)	10%	70.58	50.96	65.98	96.75	98.37	96.38	95.18	92.32	93.60
	15%	62.02	45.40	55.68	99.54	99.82	99.48	97.57	96.10	96.77
	20%	80.83	66.18	77.95	99.96	99.98	99.96	99.15	98.57	98.88
SpectralNet	5%	31.20	16.74	22.46	99.48	99.75	99.43	97.91	96.34	97.24
Chakraborty and Trehan (2021)	10%	94.03	75.96	93.20	99.73	99.83	99.70	98.99	98.16	98.67
	15%	97.65	86.10	97.32	99.97	99.97	99.97	99.62	99.40	99.49
	20%	98.51	98.34	98.30	99.99	99.98	99.99	99.70	99.52	99.60
HybridSN	5%	92.71	87.23	91.67	**99.96**	99.96	99.95	99.13	98.33	98.85
Roy *et al*. (2020)	10%	97.91	97.81	97.62	99.97	99.96	99.97	99.75	99.13	99.67
	15%	98.65	98.14	98.46	99.98	99.99	99.97	99.88	99.81	99.85
	20%	99.10	99.10	98.98	99.99	99.99	99.99	99.98	99.98	99.98
2D-3D Multibranch	5%	**92.96**	74.02	91.96	99.85	99.88	99.83	99.39	98.80	99.19
Ge *et al*. (2020)	10%	98.18	90.45	97.93	99.95	99.85	99.94	**99.76**	99.42	99.68
	15%	98.51	98.44	98.31	99.99	99.98	99.99	99.91	99.86	99.89
	20%	98.96	93.58	98.81	99.98	99.99	99.98	99.94	99.87	99.91
Proposed Method	5%	90.14	71.75	88.70	99.95	99.93	99.91	**99.40**	98.17	98.88
	10%	**98.20**	97.43	97.77	**99.99**	99.97	99.99	99.71	98.95	99.44
	15%	**98.91**	98.82	98.53	**99.99**	99.99	99.99	**99.96**	99.80	99.79
	20%	**99.27**	99.40	99.14	**100.00**	100.00	100.00	**99.99**	99.88	99.95

respectively. Similarly, for the SA dataset, the proposed method achieves an OA of 100% compared to HybridSN and 2D-3D Multibranch models with OAs of 99.92% and 99.93%, respectively. For the PU dataset, the proposed method achieves an OA of 99.95%, outperforming HybridSN and 2D-3D Multibranch models with OAs of 99.90% and 99.89%, respectively. These results highlight the significant improvement in classification accuracy achieved by the proposed method compared to the HybridSN and 2D-3D Multibranch models under the same spectral features and training sample conditions.

In addition to numeric results, we compare the results graphically using three different groups representing three distinct sub-figures to provide a more lucid comparison of our proposed method with state-of-the-art approaches and enhance the visual representation. We compare the classification performance in OAs among our proposed methods, with original data + SVM and Seg-PCA + SVM in the first group, 2D-CNN, Fast 3D-CNN, and SpectralNet in the second group, and finally HybridSN and 2D-3D Multi-branch in the third group. The classification accuracy comparison is depicted in Figures 4–6 respectively, for IP, SA, and PU HSI dataset. In the first group of results, it is evident that the proposed methods exhibit significantly better performance compared to the SVM on original data and the Seg-PCA + SVM methods for all three datasets. Similarly, in the second group of

Table 6. Class-wise classification accuracies (%) of IP with 10 spectral features and 20% training samples.

Classes	Original data+SVM	SegPCA +SVM	2D-CNN	Fast 3D-CNN	SpectralNet	HybridSN	2D-3D Multibranch	Proposed Method
Alfalfa	0.00	42.56	92.50	100	100	92.75	100	86.75
Corn-notill	43.25	65.37	85.71	65.80	99.73	98.67	99.67	99.33
Corn-mintill	14.58	70.62	83.44	79.38	97.21	99.13	96.78	100
Corn	0.00	57.84	93.75	85.71	99.48	99.07	98.87	100
Grass-pasture	8.12	88.79	95.21	84.99	100	100	97.07	100
Grass-trees	47.37	87.20	87.81	79.67	99.83	97.85	98.17	100
Grass-pasture-mowed	0.00	71.00	95.65	100	100	100	99.33	100
Hay-windrowed	31.12	90.87	90.09	90.28	99.74	100	100	100
Oats	0.00	31.00	76.19	100	0.00	93.96	100	100
Soybean-notill	49.50	69.50	79.31	77.51	98.73	99.17	99.07	100
Soybean-mintill	39.37	68.75	89.53	81.38	97.61	98.81	98.78	98.93
Soybean-clean	0.00	67.87	80.28	69.52	98.52	99.23	99.69	99.69
Wheat	0.00	91.12	95.35	93.37	100	99.12	99.87	100
Woods	72.25	94.33	97.82	98.56	99.61	100	100	100
Buildings-GTD	0.00	70.57	68.47	85.04	98.72	99.87	99.93	100
Stone-Steel-Towers	100	98.08	91.36	88.89	97.10	92.76	93.37	92.67
OA	46.44	75.55	87.04	80.83	98.51	99.10	98.96	99.27
AA	26.89	67.36	93.60	66.18	98.34	99.10	93.58	99.40
Kappa	35.41	71.78	85.35	77.95	98.30	98.98	98.81	99.14

Table 7. Class-wise classification accuracies (%) of SA with 10 spectral features and 20% training samples.

Classes	Original data+SVM	SegPCA +SVM	2D-CNN	Fast 3D-CNN	SpectralNet	HybridSN	2D-3D Multibranch	Proposed Method
Brocoli-green-weeds-1	91.71	100	100	100	100	100	100	100
Brocoli-green-weeds-2	90.28	100	99.90	100	100	100	100	100
Fallow	53.23	97.07	100	100	100	100	100	100
Fallow-rough-plow	61.55	98.74	99.64	99.91	100	99.07	98.13	100
Fallow-smooth	60.21	99.14	100	100	100	100	99.67	100
Stubble	96.68	100	99.97	100	100	100	100	100
Celery	90.16	100	100	100	100	100	100	100
Grapes-untrained	57.56	78.52	98.86	99.91	98.74	100	100	100
Soil-vinyard-develop	82.34	99.58	100	100	100	100	100	100
Corn-senesced-GW	60.10	96.85	100	99.96	100	100	100	100
Lettuce-romaine-4wk	0.00	98.63	100	100	100	100	100	100
Lettuce-romaine-5wk	83.03	98.17	100	100	100	100	100	100
Lettuce-romaine-6wk	68.08	98.67	100	100	100	100	100	100
Lettuce-romaine-7wk	66.42	98.14	100	100	100	99.83	100	100
Vinyard-untrained	54.53	82.50	90.50	99.91	99.16	100	99.40	100
Vinyard-vertical-trellis	82.57	100	100	100	100	100	100	100
OA	71.74	92.71	98.37	99.96	99.99	99.99	99.98	100
AA	68.46	96.13	99.44	99.98	99.98	99.99	99.99	100
Kappa	68.13	91.87	98.19	99.96	99.99	99.99	99.98	100

results, the proposed methods clearly outperform the 2D-CNN, Fast 3D-CNN, and SpectralNet methods for all three datasets. Moving on to the third group of results, the proposed methods demonstrate a notable improvement compared to the HybridSN and 2D-3D Multi-branch methods, albeit to a lesser extent. Overall, the results consistently indicate that the proposed methods outperform or surpass the performance of several studied benchmark methods across all three datasets. As training samples increase, all models exhibit improved performance. This indicates that a larger training dataset allows the models to learn more representative and discriminative features, enhancing classification accuracy.

Table 8. Class-wise classification accuracies (%) of PU with 10 spectral features and 20% training samples.

Classes	Original data+SVM	SegPCA +SVM	2D-CNN	Fast 3D-CNN	SpectralNet	HybridSN	2D-3D Multibranch	Proposed Method
Asphalt	53.96	84.76	99.70	99.05	100	100	100	100
Meadows	68.47	92.57	99.92	99.67	100	100	100	100
Gravel	39.54	83.45	97.54	98.98	98.78	99.93	99.74	98.37
Trees	0.00	96.10	98.65	98.63	99.58	99.75	98.37	100
Painted MS	94.67	100	100	99.72	100	100	100	100
Bare Soil	40.78	92.84	99.80	99.73	100	100	100	100
Bitumen	0.00	93.07	98.70	98.02	99.18	99.87	100	100
Self-BB	47.18	80.58	97.43	97.10	98.82	100	98.54	100
Shadows	0.00	100	99.47	97.75	100	100	100	100
OA	62.62	90.61	99.38	99.15	99.70	99.98	99.94	99.99
AA	38.53	83.82	99.07	98.57	99.52	99.98	99.87	99.88
Kappa	45.56	87.39	99.18	98.88	99.60	99.98	99.91	99.95

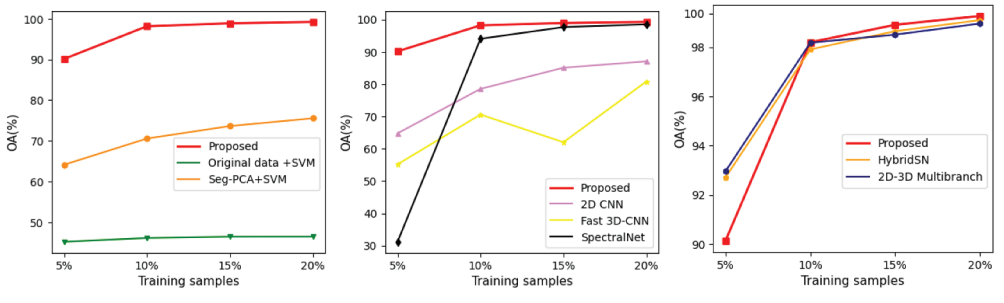


Figure 4. Classification accuracies comparison of IP data among the proposed methods with (a) original data + SVM, Seg-PCA+SVM, (b) 2D-CNN, fast 3D-CNN, SpectralNet and (c) HybridSN, 2D-3D multi-branch, using 10 spectral bands and the different number of training samples.

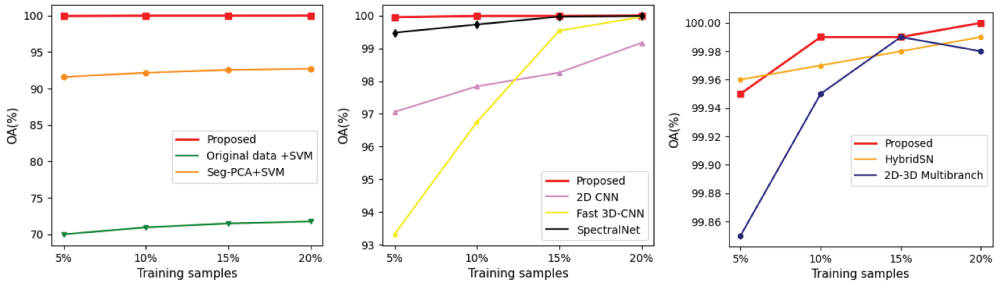


Figure 5. Classification accuracies comparison of SA data among the proposed methods with (a) original data + SVM, Seg-PCA+SVM, (b) 2D-CNN, fast 3D-CNN, SpectralNet and (c) HybridSN, 2D-3D multi-branch, using 10 spectral bands and the different number of training samples.

The convergence of accuracy and loss for the proposed methods is illustrated in Figure 7 during 100 epochs of training and validation. The results demonstrate that convergence is attained around the 45th epoch, indicating a relatively rapid convergence

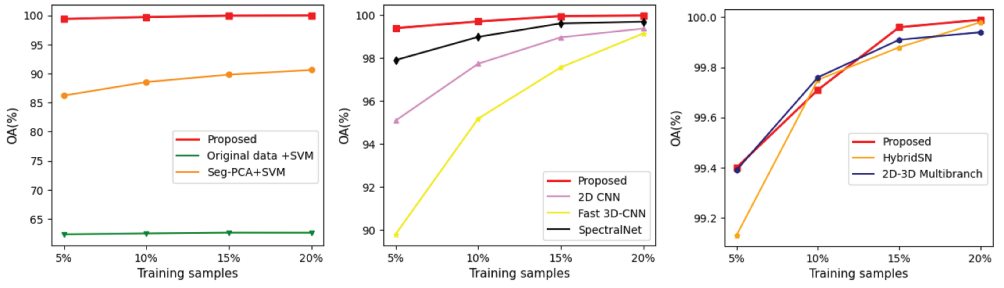


Figure 6. Classification accuracies comparison of PU data among the proposed methods with (a) original data + SVM, Seg-PCA+SVM, (b) 2D-CNN, fast 3D-CNN, SpectralNet and (c) HybridSN, 2D-3D multi-branch, using 10 spectral bands and the different number of training samples.

rate for our method. This observation highlights the effectiveness and efficiency of our proposed approach in capturing the underlying patterns of the data, leading to improved classification performance.

Our proposed model distinguishes itself through a pixel-level fusion of spectral and spatial information using a multibranch 3D-2D CNN, contributing to enhanced accuracy. In contrast, object-oriented image analysis (OOIA) operates at a higher level by grouping pixels into objects based on spatial properties, engaging in contextual classification of these objects. The nuanced approach of our model involves a detailed analysis of individual pixel characteristics, providing fine-grained details for specific hyperspectral image classification tasks.[1]

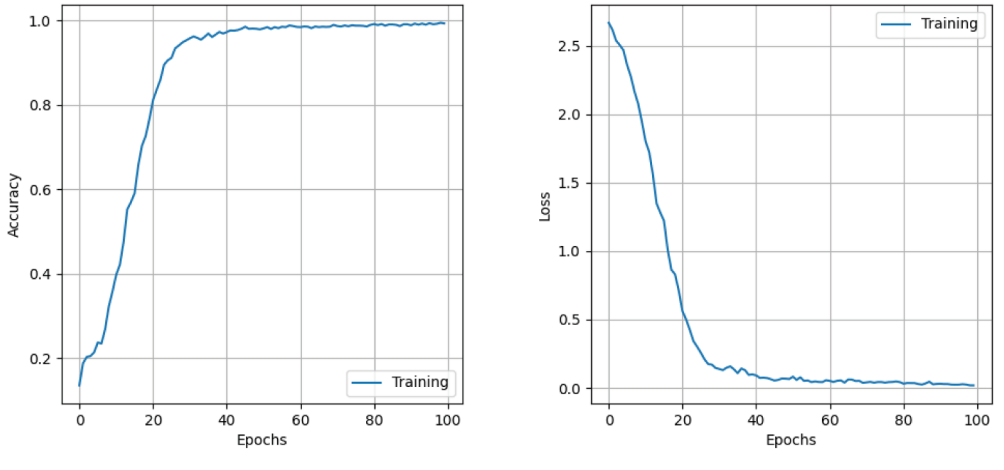


Figure 7. IP data, accuracy, and loss convergence versus the number of epochs over 5 spectral bands and 20% training samples.

4.5. Classification maps

Besides the quantitative evaluation, a visual assessment of classification maps is conducted to validate the effectiveness of the proposed method. Figures 8–10 illustrate the classification maps obtained by different methods for different HSIs. The maps are generated in 10% and 20% training samples for the proposed and all studied methods. It is evident from the visual inspection that the proposed method outperforms the other methods across all datasets. The classification maps generated by the proposed method exhibit higher clarity and fewer misclassified pixels, indicating its superior performance in capturing the complex spatial and spectral information present in hyperspectral imagery.

4.6. Ablation study

We created three distinct ablation experiments to demonstrate the efficacy of the proposed Seg-PCA, 3D-2D CNN, and multi-branch feature fusion components. The network of the proposed method is consistent with the models utilised for comparison. First, the Seg-PCA module's PCA preprocessing was removed to determine its effectiveness. Next, only 3D-CNN were used in the multi-branch stages to understand the necessity of combining 3D-2D CNN. Finally, only one branch of 3D-2D CNN was employed, and the other branches were eliminated to evaluate the efficiency of the multi-branch feature fusion. In our ablation experiments, we utilise two different training-testing splits, where the training samples account for 10% and 20% of the dataset, respectively. Moreover, we consider 5 spectral bands or features for the ablation studies.

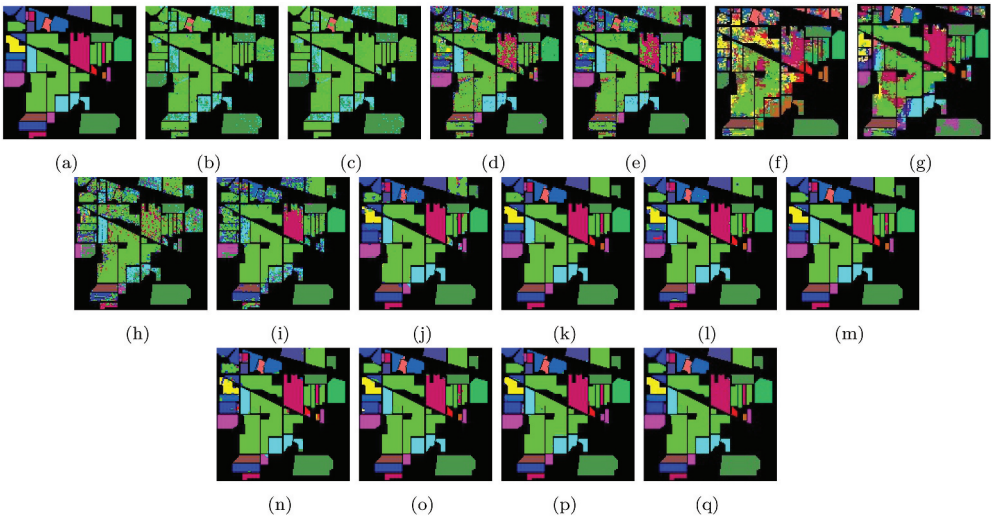


Figure 8. IP HSI (a) ground truth image, (b)-(q) classification maps for different SVM and CNN model with 5 spectral features, (b) original data + SVM (10%), (c) original data + SVM (20%), (d) Seg-PCA + SVM (10%), (e) Seg-PCA + SVM (20%), (f) 2D CNN (10%), (g) 2D CNN (20%), (h) fast 3D-CNN (10%), (i) fast 3D-CNN (20%), (j) SpectralNet (10%), (k) SpectralNet (20%), (l) HybridSN (10%), (m) HybridSN (20%), (n) 2D-3D multibranch (10%), (o) 2D-3D multibranch (20%), (p) proposed method (10%), and (q) proposed method (20%). Note that 10% and 20% are the % of the training sample.

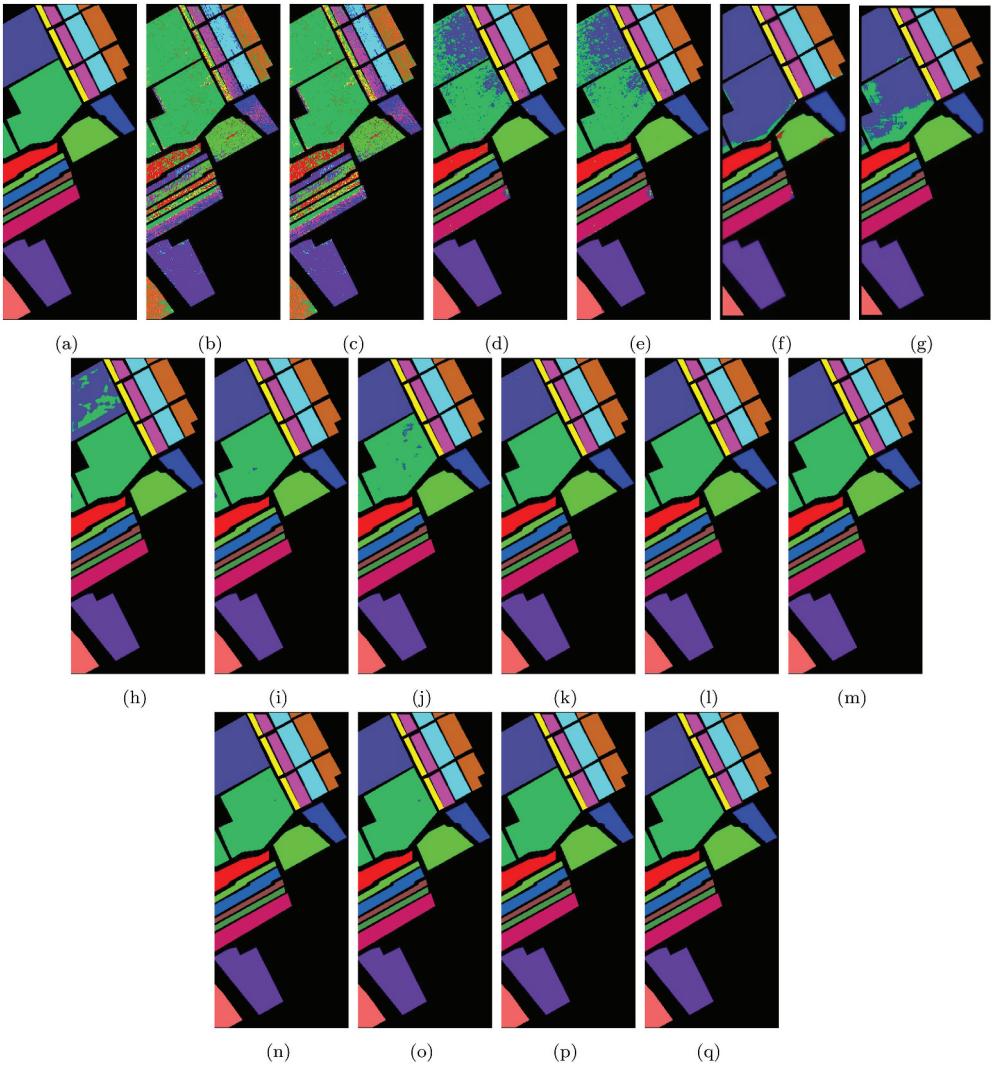


Figure 9. SA HSI (a) ground truth image, (b)-(q) classification maps for different SVM and CNN model with 5 spectral features, (b) original data + SVM (10%), (c) original data + SVM (20%), (d) Seg-PCA + SVM (10%), (e) Seg-PCA + SVM (20%), (f) 2D CNN (10%), (g) 2D CNN (20%), (h) fast 3D-CNN (10%), (i) fast 3D-CNN (20%), (j) SpectralNet (10%), (k) SpectralNet (20%), (l) HybridSN (10%), (m) HybridSN (20%), (n) 2D-3D multibranch (10%), (o) 2D-3D multibranch (20%), (p) proposed method (10%), and (q) proposed method (20%). Note that 10% and 20% are the % of the training sample.

The classification results for the three different ablation experiments with 5 spectral features are summarised in Table 9.

4.6.1. Removal of PCA prepossessing in Seg-PCA

In this ablation experiment, we perform segmentation on the dataset, dividing it into three segments. From each segment, we select the first 5 spectral bands as input for the three different 3D-2D CNN branches. Prior to applying the 3D-2D CNN, the data is

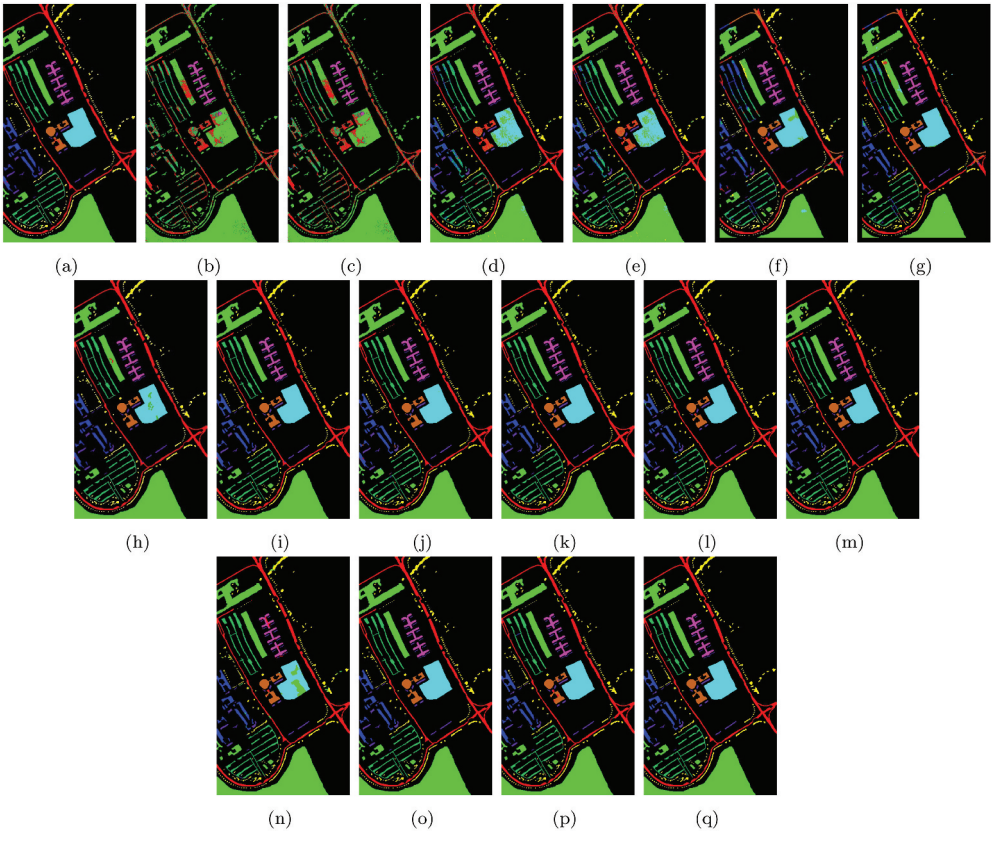


Figure 10. PU HSI (a) ground truth image, (b)-(q) classification maps for different SVM and CNN model with 5 spectral features, (b) original data + SVM (10%), (c) original data + SVM (20%), (d) Seg-PCA + SVM (10%), (e) Seg-PCA + SVM (20%), (f) 2D CNN (10%), (g) 2D CNN (20%), (h) fast 3D-CNN (10%), (i) fast 3D-CNN (20%), (j) SpectralNet (10%), (k) SpectralNet (20%), (l) HybridSN (10%), (m) HybridSN (20%), (n) 2D-3D multibranch (10%), (o) 2D-3D multibranch (20%), (p) proposed method (10%), and (q) proposed method (20%). Note that 10% and 20% are the % of the training sample.

Table 9. Classification results for ablation experiments with 5 spectral features.

Method	Training Samples	IP			SA			PU		
		OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
Removal of PCA prepossessing	10%	94.25	72.45	74.86	98.70	99.44	98.55	98.61	98.03	98.16
	20%	96.23	86.04	95.69	93.07	95.70	92.31	99.70	99.48	99.61
Removal of 2D-CNN (3D-CNN only)	10%	95.14	79.40	94.44	99.84	99.84	99.82	99.47	99.02	99.30
	20%	98.95	96.65	98.91	99.65	99.83	99.61	99.71	99.44	99.62
Removal of 2 branches (Single 3D-2D branch)	10%	95.89	83.25	95.30	99.88	99.86	99.87	98.29	97.24	97.74
	20%	98.35	91.54	98.12	99.92	99.95	99.91	99.77	99.73	99.70
Proposed Method	10%	97.05	82.63	96.52	99.99	99.97	99.98	99.78	99.31	99.63
	20%	99.07	98.94	98.46	100.00	99.99	100.00	99.95	99.88	99.89

preprocessed using min-max scaling. Subsequently, the remaining steps are the same as in the original proposed method. Table 9 presents the results of the ablation study without PCA for the three datasets. It is evident that the proposed method outperforms the without PCA ablation in terms of all three classification performance metrics: OA, AA,

and Kappa. The PCA ablation yields OA values of 94.25% and 96.23%, 98.70% and 93.07%, and 98.61% and 99.70% for IP, SA and PU HSIs, respectively, with 10% and 20% training samples. In comparison, the proposed method achieves OA values of 97.05% and 99.07%, 99.99% and 100%, and 99.95% and 99.88% for IP, SA and PU HSIs, respectively, with the same training samples. Therefore, the proposed method outperforms the PCA ablation regarding classification accuracy.

4.6.2. Removal of 2D-CNN from 3D-2D CNN

In the second ablation experiment, we removed the 2D-CNN layer from each 3D-2D branch to isolate the effect of the 3D-CNN component alone. By doing so, we aimed to evaluate the impact of the 3D-CNN in isolation or, conversely, the combined effect of the 3D-2D CNN. All other components of our proposed model remained the same in this experiment.

Based on the ablation analysis results, it appears that the proposed method, which includes all components, including the 2D-CNN, achieves higher classification accuracy compared to the ablation method, where the 2D-CNN is removed. The results shown in Table 9 demonstrate that the removal of the 2D-CNN leads to lower OA values for all three types of HSIs (IP, SA, and PU) and with different percentages of training samples. For example, when 10% of training samples are used, the removal of the 2D-CNN, results in OA values of 95.14%, 99.84%, and 99.47% for IP, SA, and PU HSIs, respectively. In contrast, the proposed method achieves higher OA values of 97.05%, 99.99%, and 99.78% for the same types of HSIs. Similarly, when 20% of training samples are used, the removal of the 2D-CNN leads to OA values of 98.95%, 99.65%, and 99.71% for IP, SA, and PU HSIs, respectively. On the other hand, the proposed method achieves higher OA values of 99.07%, 100%, and 99.88% for the same types of HSIs. Based on these results, it can be concluded that the proposed method, which includes the 2D-CNN, outperforms the ablation method without the 2D-CNN in terms of classification accuracy for all three types of HSIs.

4.6.3. Single 3D-2D CNN branch

In this ablation study, we perform the following steps removing two branches or taking a single 3D-2D CNN branch. First, we apply Seg-PCA and min-max scaling to the data. Then, we concatenate the PCs obtained from each segment. Then we select the top-ranked 5 features based on their variances. The selected fused feature is passed through a single 3D-2D CNN branch. The remaining parts of the classification process remain the same as in our proposed method.

Based on the information from the ablation study, it appears that the proposed method, which includes all components, including the three 3D-2D CNN branches, achieves higher classification accuracy compared to the ablation method, which uses a single 3D-2D CNN branch. The results presented in Table 9 reveal that removing two branches leads to decreased OA values for all three categories of HSIs, namely IP, SA, and PU, across varying percentages of training samples. For instance, when utilising 10% of the training samples, employing a single 3D-2D CNN branch yields OA values of 95.89%, 99.88%, and 98.29% for IP, SA, and PU HSIs, respectively. Conversely, the proposed method, encompassing three 3D-2D CNN branches, achieves higher OA values of 97.05%, 99.99%, and 99.78% for the corresponding types of HSIs. Likewise, when utilising 20% of the training samples, the

usage of a single 3D-2D CNN branch results in OA values of 98.35%, 99.92%, and 99.77% for IP, SA, and PU HSIs, respectively. Conversely, the proposed method attains higher OA values of 99.07%, 100%, and 99.88% for the respective HSIs.

Based on these findings, it can be concluded that the proposed method, which incorporates three 3D-2D CNN branches, surpasses the ablation method featuring only a single 3D-2D CNN branch in terms of classification accuracy across all three types of HSIs.

5. Conclusion and future work

HSI classification has gained significant attention from remote-sensing researchers due to its capability to identify objects from long distances. Various machine learning algorithms have been previously evaluated for efficient HSI classification. However, some of these algorithms have limitations in terms of accuracy, while others are associated with high computational costs. Deep CNN architectures have recently shown superior performance in HSI classification. This article introduces a novel HSI classification approach integrating Seg-PCA and a 3D-2D CNN with multi-branch feature fusion. Our method addresses the existing challenges in HSI classification by optimising the formation of the CNN-based model and establishing a multi-branch composition. The proposed model takes the data through multiple 3D-2D CNN branches and manages different local PCA features extracted from distinct subgroups of data, fuse the extracted features from these branches and produce the classification results through three fully connected layers and a softmax output layer. Experimental results demonstrate that the proposed method exhibits exceptional performance in terms of OA, AA, and Kappa. Comparative experiments conducted on three benchmark datasets against state-of-the-art methods confirm the superiority of our proposed method. Furthermore, the proposed method exhibits favourable classification results even with a small amount of training sample, showcasing its effectiveness.

The proposed method has limitations, including slightly lower accuracy in the IP dataset than SA and PU. The neural network's fully connected layers have too many parameters. Future work aims to create a more compact model for better generalisation and performance across datasets. Additionally, PCA transforms with a sliding window on 2D images could adaptively reduce data dimensionality, improving classification performance.

Note

1. Our work uniquely focuses on pixel-level enhancement in HSI analysis, complementing the relevance of OOIA in varied contexts. Future research could explore synergies and integrate these approaches for a more comprehensive HSI analysis.

Disclosure statement

We hereby declare that there are no known conflicts of interest associated with this paper. All named authors have read and approved the paper, and there are no other individuals who meet the criteria for authorship but have not been included. All have unanimously approved the order of authors listed in the paper.

References

Afjal, M.I., Mondal, M.N.I., and Mamun, M.A., 2022. Segmented linear discriminant analysis for hyperspectral image classification. *In*: *IEEE 12th international conference on electrical and computer engineering (ICECE)*. Dhaka, Bangladesh: IEEE, 1–4.

Afjal, M.I., Mondal, M.N.I., and Mamun, M.A., 2023. Segmentation-based linear discriminant analysis with information theoretic feature selection for hyperspectral image classification. *International Journal of Remote Sensing*, 44 (11), 3412–3455. doi:10.1080/01431161.2023.2221802

Ahmad, M., *et al*., 2020. A fast and compact 3-d cnn for hyperspectral image classification. *IEEE Geoscience and Remote Sensing Letters*, 19, 1–5.

Aloysius, N. and Geetha, M., 2017. A review on deep convolutional neural networks. *In*: *2017 international conference on communication and signal processing (ICCSP)*. Chennai, India: IEEE, 0588–0592.

Balaji, K., *et al*., 2023. DesU-NetAM: optimized DenseU-net with attention mechanism for hyperspectral image classification. *International Journal of Information Technology*, 15 (7), 3761–3777. doi:10.1007/s41870-023-01386-5.

Bandos, T.V., Bruzzone, L., and Camps-Valls, G., 2009. Classification of hyperspectral images with regularized linear discriminant analysis. *IEEE Transactions on Geoscience and Remote Sensing*, 47 (3), 862–873. doi:10.1109/TGRS.2008.2005729

Baumgardner, M.F., Biehl, L.L., and Landgrebe, D.A. 2015, September. 220 band aviris hyperspectral image data set: June 12, 1992 Indian pine test site 3.

Ben Hamida, A., *et al*., 2018. 3-d deep learning approach for remote sensing image classification. *IEEE Transactions on Geoscience and Remote Sensing*, 56 (8), 4420–4434. doi:10.1109/TGRS.2018.2818945.

Chakraborty, T. and Trehan, U., 2021. SpectralNET: exploring spatial-spectral WaveletCNN for hyperspectral image classification. *arXiv*.

Chang, C.C. and Lin, C.J., 2011. Libsvm: a library for support vector machines. *ACM Transactions on Intelligent Systems and Technology*, 2 (3), 1–27. doi:10.1145/1961189.1961199

Chen, H., *et al*., 2023. Hyperspectral image classification based on fusing S3-PCA, 2D-SSA and random patch network. *Remote Sensing (Basel)*, 15 (13), 3402. doi:10.3390/rs15133402.

Chen, Y., Zhao, X., and Jia, X., 2015. Spectral–spatial classification of hyperspectral data based on deep belief network. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 8 (6), 2381–2392. doi:10.1109/JSTARS.2015.2388577

Clark, A., Phinn, S., and Scarth, P., 2023. Optimised U-Net for land use–land cover classification using aerial photography. *PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science*, 91 (2), 125–147. doi:10.1007/s41064-023-00233-3

Fabiyi, S.D., *et al*., 2021. Folded lda: extending the linear discriminant analysis algorithm for feature extraction and data reduction in hyperspectral remote sensing. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 14, 12312–12331. doi:10.1109/JSTARS.2021.3129818.

Fang, L., *et al*., 2015. Classification of hyperspectral images by exploiting spectral–spatial information of superpixel via multiple kernels. *IEEE Transactions on Geoscience and Remote Sensing*, 53 (12), 6663–6674. doi:10.1109/TGRS.2015.2445767.

Gamba, P., 2004. A collection of data for urban area characterization. *In*: *IGARSS 2004. 2004 IEEE international geoscience and remote sensing symposium*. Vol. 1. Anchorage, AK: IEEE, 72.

Ge, Z., *et al*., 2020. Hyperspectral image classification method based on 2d–3d cnn and multibranch feature fusion. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 13, 5776–5788. doi:10.1109/JSTARS.2020.3024841.

GIC, 2021. Hyperspectral remote sensing scenes, salinas. Available from: https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes [Accessed 15 July 2021].

Guo, A.J.X. and Zhu, F., 2019. A cnn-based spatial feature fusion algorithm for hyperspectral imagery classification. *IEEE Transactions on Geoscience and Remote Sensing*, 57 (9), 7170–7181. doi:10.1109/TGRS.2019.2911993

Hajaj, S., *et al*., 2023. HyMap imagery for copper and manganese prospecting in the east of ameln valley shear zone (kerdous inlier, western anti-atlas, morocco). *Journal of Spatial Science*, 1–22. 10.1080/14498596.2023.2172085.

Hassanzadeh, S., Danyali, H., and Helfroush, M.S., 2022. Hyperspectral images classification based on multiple kernel learning using swt and kmnf with few training samples. *Journal of Spatial Science*, 64 (4), 593–613.

He, K., *et al*., 2016. Deep residual learning for image recognition. *In: 2016 IEEE conference on computer vision and pattern recognition (CVPR)*. Las Vegas, NV: IEEE, 770–778.

He, N., *et al*., 2019. Feature extraction with multiscale covariance maps for hyperspectral image classification. *IEEE Transactions on Geoscience and Remote Sensing*, 57 (2), 755–769. doi:10.1109/TGRS.2018.2860464.

Hell, M., *et al*., 2022. Classification of tree species and standing dead trees with lidar point clouds using two deep neural networks: PointCNN and 3DmFV-net. *PFG –Journal of Photogrammetry Remote Sensing Geoinformation Science*, 90 (2), 103–121. doi:10.1007/s41064-022-00200-4.

Hong, D., *et al*., 2021. Graph convolutional networks for hyperspectral image classification. *IEEE Transactions on Geoscience and Remote Sensing*, 59 (7), 5966–5978. doi:10.1109/TGRS.2020.3015157.

Hughes, G., 1968. On the mean accuracy of statistical pattern recognizers. *IEEE Transactions on Information Theory*, 14 (1), 55–63. doi:10.1109/TIT.1968.1054102

Islam, R., Ahmed, B., and Hossain, A., 2022. Feature reduction of hyperspectral image for classification. *Journal of Spatial Science*, 67 (2), 331–351. doi:10.1080/14498596.2020.1770137

Islam, M.R., Islam, M.T., and Uddin, M.P., 2023. Improving hyperspectral image classification through spectral-spatial feature reduction with a hybrid approach and deep learning. *Journal of Spatial Science*, 1–18. doi:10.1080/14498596.2023.2227948.

Jia, X. and Richards, J.A., 1999. Segmented principal components transformation for efficient hyperspectral remote-sensing image display and classification. *IEEE Transactions on Geoscience and Remote Sensing*, 37 (1), 538–542. doi:10.1109/36.739109

Joelsson, S., Benediktsson, J., and Sveinsson, J., 2005. Random forest classifiers for hyperspectral data. *In: Proceedings. 2005 IEEE international geoscience and remote sensing symposium, 2005. IGARSS '05*. Vol. 1, 4.

Lei, Z., *et al*., 2022. Active deep learning for hyperspectral image classification with uncertainty learning. *IEEE Geoscience and Remote Sensing Letters*, 19, 1–5. doi:10.1109/LGRS.2022.3204344.

Li, M., *et al*., 2022. Automatic assessment of depression and anxiety through encoding pupil-wave from HCI in VR scenes. *ACM Transactions on Multimedia Computing, Communications and Applications*, 20 (2), 1–22. doi:10.1145/3513263.

Li, R., *et al*., 2020. Classification of hyperspectral image based on double-branch dual-attention mechanism network. *Remote Sens (Basel)*, 12 (3), 582. doi:10.3390/rs12030582.

Li, S., *et al*., 2019. Deep learning for hyperspectral image classification: an overview. *IEEE Transactions on Geoscience and Remote Sensing*, 57 (9), 6690–6709. doi:10.1109/TGRS.2019.2907932.

Li, W., *et al*., 2017. Hyperspectral image classification using deep pixel-pair features. *IEEE Transactions on Geoscience and Remote Sensing*, 55 (2), 844–853. doi:10.1109/TGRS.2016.2616355.

Li, Y., *et al*., 2018. Deep learning for remote sensing image classification: a survey. *Wiley Interdisciplinary Reviews Data Mining and Knowledge Discovery*, 8 (6), e1264. doi:10.1002/widm.1264.

Liang, F., *et al*., 2018. Hyperspectral image recognition based on artificial neural network. *Neuroquantology*, 16 (5). doi:10.14704/nq.2018.16.5.1244.

Liu, H., *et al*., 2021. Deep fuzzy graph convolutional networks for polsar imagery pixelwise classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 14, 504–514. doi:10.1109/JSTARS.2020.3041534.

Liu, J., *et al*., 2023. Adaptive multi-feature fusion graph convolutional network for hyperspectral image classification. *Remote Sensing (Basel)*, 15 (23), 5483. doi:10.3390/rs15235483.

Makantasis, K., *et al*., 2015. Deep supervised learning for hyperspectral data classification through convolutional neural networks. *In: 2015 IEEE international geoscience and remote sensing symposium (IGARSS)*. Milan, Italy: IEEE, 4959–4962.

Nega, W., Nigussie, W., and Balew, A., 2022. Investigating spatial relationship between vegetation cover and soil moisture using sentinel satellites: a case study of north Gondar zone, Ethiopia. *Journal of Spatial Science*, 68 (4), 667–683. doi:10.1080/14498596.2022.2133019

Nickolls, J. and Dally, W.J., 2010. The gpu computing era. *IEEE Micro*, 30 (2), 56–69. doi:10.1109/MM.2010.41

Pan, B., Shi, Z., and Xu, X., 2018. Mugnet: deep learning for hyperspectral image classification using limited samples. *ISPRS Journal of Photogrammetry and Remote Sensing*, 145, 108–119. Deep Learning RS Data.

Paul, A., Bhattacharyya, S., and Chakraborty, D., 2021. Estimation of shade tree density in tea garden using remote sensing images and deep convolutional neural network. *Journal of Spatial Science*, 68 (3), 415–429. doi:10.1080/14498596.2021.2013966

Pierdicca, R., *et al*., 2020. Point cloud semantic segmentation using a deep learning framework for cultural heritage. *Remote Sens (Basel)*, 12 (6), 1005. doi:10.3390/rs12061005.

Prasad, S. and Bruce, L.M., 2008. Limitations of principal components analysis for hyperspectral target recognition. *IEEE Geoscience and Remote Sensing Letters*, 5 (4), 625–629. doi:10.1109/LGRS.2008.2001282

Rodarmel, C. and Shan, J., 2002. Principal component analysis for hyperspectral image classification. *Surv Land Inf Syst*, 62 (2), 115–123.

Roy, S.K., *et al*., 2020. Hybridsn: exploring 3-d–2-d cnn feature hierarchy for hyperspectral image classification. *IEEE Geoscience and Remote Sensing Letters*, 17 (2), 277–281. doi:10.1109/LGRS.2019.2918719.

Sen, O. and Keles, H.Y., 2022. A hierarchical approach to remote sensing scene classification. *PFG – Journal of Photogrammetry Remote Sensing Geoinformation Science*, 90 (2), 161–175. doi:10.1007/s41064-022-00193-0

Siddiqa, A., Islam, R., and Afjal, M.I., 2022. Spectral segmentation based dimension reduction for hyperspectral image classification. *Journal of Spatial Science*, 68 (4), 543–562. doi:10.1080/14498596.2022.2074902

Song, W., *et al*., 2016. Hyperspectral image classification based on knn sparse representation. *In: 2016 IEEE international geoscience and remote sensing symposium (IGARSS)*. Beijing, China: IEEE, 2411–2414.

Tsai, F., Lin, E., and Yoshino, K., 2007. Spectrally segmented principal component analysis of hyperspectral imagery for mapping invasive plant species. *International Journal of Remote Sensing*, 28 (5), 1023–1039. doi:10.1080/01431160600887706

Uddin, M.P., *et al*., 2021a. Information-theoretic feature selection with segmentation-based folded principal component analysis (pca) for hyperspectral image classification. *International Journal of Remote Sensing*, 42 (1), 286–321.

Uddin, M.P., *et al*., 2022. Improved folded-pca for efficient remote sensing hyperspectral image classification. *Geocarto International*, 37 (25), 9474–9496. doi:10.1080/10106049.2021.2020343.

Uddin, M.P., Mamun, M.A., and Hossain, M.A., 2021b. Pca-based feature reduction for hyperspectral remote sensing image classification. *IETE Technical Review*, 38 (4), 377–396.

Wang, J. and Chang, C.I., 2006. Independent component analysis-based dimensionality reduction with applications in hyperspectral image analysis. *IEEE Transactions on Geoscience and Remote Sensing*, 44 (6), 1586–1600. doi:10.1109/TGRS.2005.863297

Waske, B., *et al*., 2010. Sensitivity of support vector machines to random feature selection in classification of hyperspectral data. *IEEE Transactions on Geoscience and Remote Sensing*, 48 (7), 2880–2889. doi:10.1109/TGRS.2010.2041784.

Xie, B., *et al*., 2022. Spectral variation augmented representation for hyperspectral imagery classification with few labeled samples. *IEEE Transactions on Geoscience and Remote Sensing*, 60, 1–12. doi:10.1109/TGRS.2022.3220579.

Yang, D., *et al*., 2023. Resnet-unet considering patches (rup) network to solve the problem of patches due to shadows in extracting building top information. *Journal of Spatial Science*, 1–21. 10.1080/14498596.2023.2217782.

Yu, C., *et al*., 2022. Multiview calibrated prototype learning for few-shot hyperspectral image classification. *IEEE Transactions on Geoscience and Remote Sensing*, 60, 1–13. doi:10.1109/TGRS.2022.3225947.

Yuan, X., *et al*., 2018. Deep learning-based feature representation and its application for soft sensor modeling with variable-wise weighted sae. *IEEE Transactions on Industrial Informatics*, 14 (7), 3235–3243. doi:10.1109/TII.2018.2809730.

Zhang, D., *et al*., 2015. Weakly supervised learning for target detection in remote sensing images. *IEEE Geoscience and Remote Sensing Letters*, 12 (4), 701–705. doi:10.1109/LGRS.2014.2358994.

Zhao, H., Wu, Y., and Deng, W., 2023. An interpretable dynamic inference system based on fuzzy broad learning. *IEEE Transactions on Instrumentation and Measurement*, 72, 1–12. doi:10.1109/TIM.2023.3316213

Zhong, Z., *et al*., 2018. Spectral–spatial residual network for hyperspectral image classification: a 3-d deep learning framework. *IEEE Transactions on Geoscience and Remote Sensing*, 56 (2), 847–858. doi:10.1109/TGRS.2017.2755542.